ELASTICITY — ASSIGNMENT 01

Single Correct · Multiple Correct · Subjective

SECTION A — Single Correct Type Q1–9

Only one option is correct.

1. The maximum load a wire can withstand without breaking, when its length is reduced to half of its original length, will
(A) be double. (B) be half. (C) be four times. (D) remain same.

2. The temperature of a wire is doubled. The Young's modulus of elasticity
(A) will also double. (B) will become four times.
(C) will remain same. (D) will decrease.

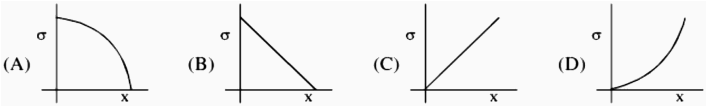
3. A spring is stretched by applying a load to its free end. The strain produced in the spring is
(A) volumetric. (B) shear.
(C) longitudinal and shear. (D) longitudinal.

4. Overall changes in volume and radii of a uniform cylindrical steel wire are 0.2% and 0.002% respectively when subjected to some suitable force. Longitudinal tensile stress acting on the wire is ($Y = 2.0 \times 10^{11}\ \mathrm{N\,m^{-2}}$)
(A) $3.2 \times 10^{9}\ \mathrm{N\,m^{-2}}$ (B) $3.2 \times 10^{7}\ \mathrm{N\,m^{-2}}$
(C) $3.6 \times 10^{9}\ \mathrm{N\,m^{-2}}$ (D) $3.9 \times 10^{8}\ \mathrm{N\,m^{-2}}$

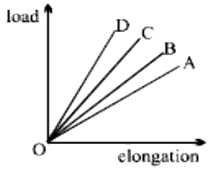
5. A solid sphere of radius R made of material of bulk modulus K is surrounded by a liquid in a cylindrical container. A massless piston of area A floats on the surface of the liquid. When a mass m is placed on the piston to compress the liquid, the fractional change in the radius of the sphere $\delta R/R$ is
(A) mg/AK (B) $mg/3AK$ (C) mg/A (D) $mg/3AR$

6. A wire suspended vertically from one of its ends is stretched by attaching a weight of 200 N to the lower end. The weight stretches the wire by 1 mm. Then the elastic energy stored in the wire is **[AIEEE 2003]**
(A) 0.2 J (B) 10 J (C) 20 J (D) 0.1 J

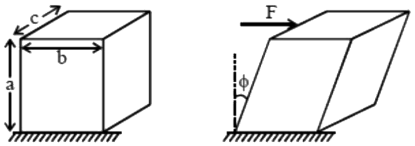
7. A uniform rod is rotating in a gravity-free region with a certain constant angular velocity. The variation of tensile stress with distance x from the axis of rotation is best represented by which of the following graphs?



8. The load versus strain graph for four wires of the same material is shown in the figure. The thickest wire is represented by the line
(A) OB (B) OA (C) OD (D) OC



9. A cuboidal block of sides a, b and c is fixed on the ground. The top is pushed by a horizontal force F as shown. The angle ϕ by which the block deforms is (η is modulus of rigidity)



(A) $\dfrac{F}{ab\eta}$ (B) $\dfrac{F}{ac\eta}$ (C) $\dfrac{F}{bc\eta}$ (D) $\dfrac{F}{\sqrt{b^2+c^2}\,\eta}$

SECTION B — Multiple Correct Type Q10–17

One or more options may be correct.

10. A wire is suspended from the ceiling and stretched under the action of a weight F suspended from its other end. The force exerted by the ceiling on it is equal and opposite to the weight.

(A) Tensile stress at any cross section A of the wire is F/A.
(B) Tensile stress at any cross section is zero.
(C) Tensile stress at any cross section A of the wire is $2F/A$.
(D) Tension at any cross section A of the wire is F.

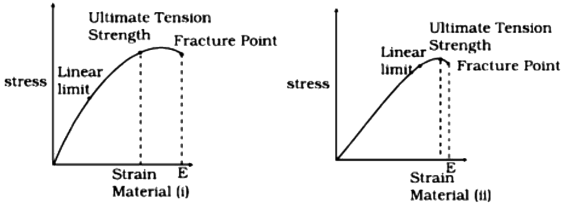
11. A copper and a steel wire of the same diameter are connected end to end. A deforming force F is applied to this composite wire which causes a total elongation of 1 cm. The two wires will have

(A) the same stress. (B) different stress.
(C) the same strain. (D) different strain.

12. A body of mass M is attached to the lower end of a metal wire, whose upper end is fixed. The elongation of the wire is l.

(A) Loss in gravitational potential energy of M is Mgl
(B) The elastic potential energy stored in the wire is Mgl
(C) The elastic potential energy stored in the wire is $\frac{1}{2}Mgl$
(D) Heat produced is $\frac{1}{2}Mgl$.

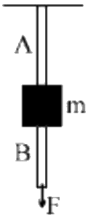
13. The stress–strain graphs for two materials are shown in the figure (assume same scale).



(A) Material (ii) is more elastic than material (i) and hence material (ii) is more brittle.
(B) Material (i) and (ii) have the same elasticity and the same brittleness.
(C) Material (ii) is elastic over a larger region of strain as compared to (i).
(D) Material (ii) is more brittle than material (i).

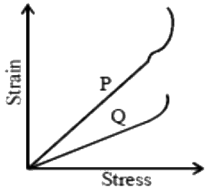
14. A composite rod consists of a steel rod of length 25 cm and area $2A$ and a copper rod of length 50 cm and area A. The composite rod is subjected to an axial load F. The Young's moduli of steel and copper are in the ratio 2 : 1.
(A) the extension produced in copper rod will be more.
(B) the extension in copper and steel parts will be in the ratio 2 : 1.
(C) the stress applied to the copper rod will be more.
(D) no extension will be produced in the steel rod.

15. The wires A and B shown in the figure are made of the same material and have radii r_A and r_B respectively. The block between them has a mass m. When the force F is $mg/3$, one of the wires breaks.



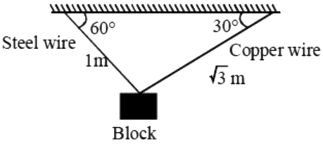
(A) A breaks if $r_A = r_B$
(B) A breaks if $r_A < 2r_B$
(C) Either A or B may break if $r_A = 2r_B$
(D) The lengths of A and B must be known to predict which wire will break

16. In plotting stress versus strain curves for two materials P and Q, a student by mistake puts strain on the y-axis and stress on the x-axis as shown in the figure. Then the correct statement(s) is (are) **[JEE Advanced 2015]**



(A) P has more tensile strength than Q
(B) P is more ductile than Q
(C) P is more brittle than Q
(D) The Young's modulus of P is more than that of Q

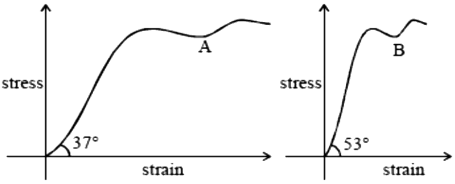
17. A block of weight 100 N is suspended by copper and steel wires of the same cross-sectional area 0.5 cm^2 and lengths $\sqrt{3}$ m and 1 m respectively. Their other ends are fixed on a ceiling as shown in the figure. The angles subtended by the copper and steel wires with the ceiling are 30° and 60° respectively. If the elongation in the copper wire is Δl_C and in the steel wire is Δl_S, then the ratio $\Delta l_C / \Delta l_S$ is ____. (Young's modulus for copper and steel: 1×10^{11} N/m^2 and 2×10^{11} N/m^2 respectively.) **[JEE Advanced 2019]**



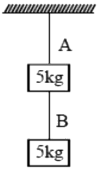
SECTION C — Subjective Questions Q18–24

18. A steel wire of length 4.5 m and a copper wire of length 3.5 m are stretched by the same amount under a given load. If the ratio of the Young's moduli of steel to that of copper is $\frac{12}{7}$, what is the ratio of the cross-sectional area of the steel wire to the copper wire?

19. The diagram shows stress–strain graphs for two materials A and B. The graphs are drawn to scale. Find the ratio of the Young's moduli of material A to material B.

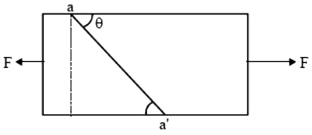


20. Two identical wires A and B of the same material are loaded as shown in the figure. If the elongation in wire B is 1.5 mm, what is the elongation in A? (Masses of A and B can be neglected.)

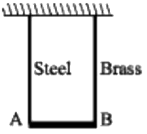


21. A wire of length L and radius r is clamped rigidly at one end. When the other end of the wire is pulled by a force f, its length increases by l. Another wire of the same material, of length $2L$ and radius $2r$, is pulled by a force $2f$. Find the increase in length of this wire.

22. Consider a long steel bar under a tensile stress due to forces F acting at the edges along the length of the bar (see figure). Consider a plane making an angle θ with the length. What are the tensile and shearing stresses on this plane? (a) For what angle is the tensile stress a maximum? (b) For what angle is the shearing stress a maximum?



23. A light rigid bar AB is suspended horizontally from two vertical wires, one of steel and one of brass, as shown in the figure. Each wire is 2.00 m long. The diameter of the steel wire is 0.60 mm and the length of the bar AB is 0.20 m. When a mass of 10 kg is suspended from the centre of AB, the bar remains horizontal. (i) What is the tension in each wire? (ii) Calculate the extension of the steel wire and the energy stored in it. (iii) Calculate the diameter of the brass wire. (iv) If the brass wire were replaced by another brass wire of diameter 1 mm, where should the mass be suspended so that AB would remain horizontal? Young's modulus for steel $= 2.0 \times 10^{11}$ Pa; for brass $= 1.0 \times 10^{11}$ Pa.



24. A wire of cross-sectional area 4×10^{-4} m^2, modulus of elasticity 2×10^{11} N/m^2 and length 1 m is stretched between two vertical rigid poles. A mass of 1 kg is suspended at its middle. Calculate the angle it makes with the horizontal.

KTG & COOLING — PRACTICE ASSIGNMENT

Pressure of an Ideal Gas · Molecular Speeds · Gas-Process Graphs · Degrees of Freedom · Newton's Law of Cooling

Use: $R = 8.3$ J mol^{-1} K^{-1} · H_2 2, He 4, N_2 28, O_2 32 g/mol (M_0 in kg/mol) · $T(\text{K}) = T(^\circ\text{C}) + 273$

SECTION A — Pressure of an Ideal Gas Q1–10

1. The temperature of a gas is raised from 27 °C to 327 °C. By what factor does (a) the rms speed of its molecules and (b) the total translational kinetic energy of the gas change?

2. Four gas molecules have speeds 1, 2, 3 and 4 km/s. Find their mean speed and rms speed. Which is larger?

3. A gas at a pressure of 1.0×10^5 Pa has a density of 1.2 kg/m^3. Find the rms speed of its molecules.

4. The pressure of an ideal gas is doubled while its temperature is kept constant. The rms speed of its molecules:
(A) doubles (B) becomes $\sqrt{2}$ times (C) becomes half (D) does not change

5. **(a)** An ideal gas fills a 2.0 L vessel at a pressure of 1.5×10^5 Pa. Find the total translational kinetic energy of its molecules. **(b)** Find the translational KE of 1 mol of any ideal gas at 300 K. ($R = 8.3$ J mol^{-1} K^{-1})

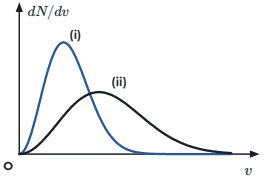
6. For nitrogen gas at 300 K ($M_0 = 28$ g/mol), find:
(a) the most probable speed v_{mp}; **(b)** the mean speed $\langle v \rangle$; **(c)** the rms speed v_{rms}; **(d)** the average **velocity** of the molecules.

7. At what temperature will the most probable speed of oxygen molecules be equal to their rms speed at 27 °C?

8. Hydrogen and oxygen are kept at the same temperature. The ratio of their mean speeds $\langle v \rangle(\text{H}_2) : \langle v \rangle(\text{O}_2)$ and the ratio of their average translational KE per molecule are respectively:
(A) $4:1$ and $1:1$ (B) $16:1$ and $1:1$
(C) $4:1$ and $1:16$ (D) $1:4$ and $1:1$

9. For a gas obeying the Maxwell–Boltzmann distribution, $v_{mp} : \langle v \rangle : v_{rms}$ is:
(A) $\sqrt{3} : \sqrt{8/\pi} : \sqrt{2}$ (B) $1:1:1$ (C) $\sqrt{2} : \sqrt{8/\pi} : \sqrt{3}$ (D) $\sqrt{2} : \sqrt{3} : \sqrt{8/\pi}$

10. The figure shows the Maxwell distribution curves (i) and (ii) for the **same amount** of the same gas at two temperatures. Which curve is at the higher temperature? Give two reasons from the graph.



SECTION B — Gas Processes & Their Graphs Q11–19

11. Match each process in Column I with the shape of its graph in Column II.

COLUMN I	COLUMN II
(A) Isothermal, on P–V	(p) Straight line through the origin
(B) Isobaric, on P–V	(q) Line parallel to the V-axis
(C) Isochoric, on P–V	(r) Rectangular hyperbola
(D) Isochoric, on P–T	(s) Line parallel to the P-axis

12. A straight line on the V–T graph of a fixed amount of gas represents an **isobaric** process only if it:

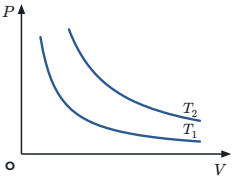
(A) has a positive slope (B) is parallel to the T-axis

(C) makes 45° with the T-axis (D) passes through the origin when extended

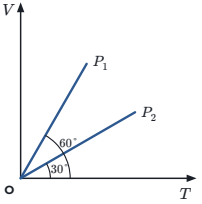
13. The figure shows two isotherms for the same amount of an ideal gas at temperatures T_1 and T_2. Then:

(A) $T_1 = T_2$ (B) $T_1 > T_2$

(C) $T_1 < T_2$ (D) cannot be decided



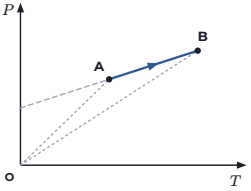
14. The V–T graphs of the same amount of an ideal gas at two constant pressures P_1 and P_2 are straight lines through the origin, making 60° and 30° with the T-axis. Find $P_1 : P_2$.



15. A fixed amount of gas goes from state A to state B along the straight line shown on the P–T graph (AB extended does **not** pass through O). During this process the volume of the gas:

(A) stays constant (B) increases

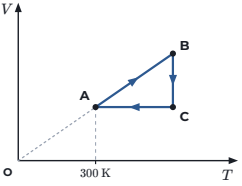
(C) decreases (D) first increases, then decreases



16. A gas goes round the cycle A → B → C → A shown on the V–T graph; AB extended passes through O. Given $T_A = 300$ K and $V_B = 2V_A$:

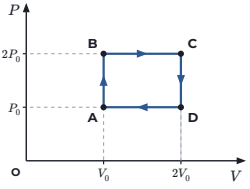
(a) name each process; **(b)** find T_B, and P_C in terms of P_A;

(c) draw the cycle on the P–V and P–T graphs.

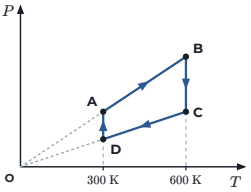


17. One mole of an ideal gas goes round the rectangular cycle A → B → C → D → A on the P–V graph. Let $T_A = T_0$.

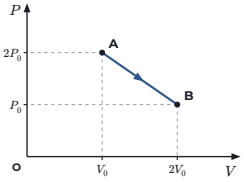
(a) name each process; **(b)** find T_B, T_C, T_D in terms of T_0. Which two states have the same temperature? **(c)** draw the cycle on the P–T and V–T graphs.



18. A gas goes round the cycle A $\to$ B $\to$ C $\to$ D $\to$ A shown on the P–T graph. AB and DC, extended, pass through O. Given $V_C = 2V_A$:
(a) name each process; **(b)** find P_B, P_C and P_D in terms of P_A; **(c)** draw the cycle on the P–V and V–T graphs.



19. An ideal gas goes from A $(V_0, 2P_0)$ to B $(2V_0, P_0)$ along the straight line shown on the P–V graph.
(a) Compare T_A and T_B. **(b)** Is the process isothermal? Explain. **(c)** Find the maximum temperature reached, in terms of T_A.



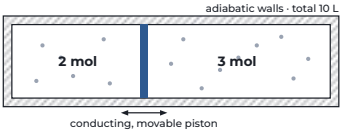
SECTION C — Degrees of Freedom & Walls Q20–23

20. Ignoring vibrations, which pair of gases has the **same** number of degrees of freedom per molecule?
(A) He and H_2 (B) N_2 and CO_2 (C) CO_2 and H_2O (D) O_2 and O_3

21. The degrees of freedom of an oxygen molecule at (i) 40 K and (ii) 6000 K are respectively:
(A) 5, 5 (B) 3, 5 (C) 5, 7 (D) 3, 7

22. A vessel contains 2 mol of helium and 3 mol of nitrogen at room temperature. Find: **(a)** the total number of degrees of freedom of all the molecules, in terms of N_A; **(b)** the average number of degrees of freedom per molecule.

23. A cylinder with adiabatic outer walls (total volume 10 L) is divided by a **conducting, freely movable** piston. The left part has 2 mol and the right part has 3 mol of an ideal gas, initially at different P and T. When everything settles:



(a) which of T, P, V become equal on the two sides? **(b)** find the final volume of each part. **(c)** if the piston were conducting but **fixed**, which quantity alone would become equal?

SECTION D — Newton's Law of Cooling Q24–30

24. A body cools in a room at 25 °C. The ratio of its rate of cooling at 60 °C to its rate of cooling at 45 °C is:
(A) 4 : 3 (B) 7 : 4 (C) 2 : 1 (D) 16 : 9

25. A body takes 4 minutes to cool from 90 °C to 80 °C in a room at 20 °C. The time it takes to cool from 80 °C to 70 °C is:
(A) less than 4 min (B) exactly 4 min
(C) more than 4 min (D) cannot be predicted

26. A liquid cools from 65 °C to 55 °C in 6 minutes in surroundings at 20 °C. Using the average-temperature method, find the time it takes to cool from 55 °C to 45 °C.

27. A body cools from 60 °C to 50 °C in 10 minutes, and from 50 °C to 42 °C in the next 10 minutes. Find the temperature of the surroundings.

28. Tea at 90 °C is left in a room at 30 °C and cools to 60 °C in 10 minutes. Using $T = T_s + (T_i - T_s)\,e^{-kt}$, find:
(a) its temperature after another 10 minutes; **(b)** the total time from the start for it to reach 37.5 °C.

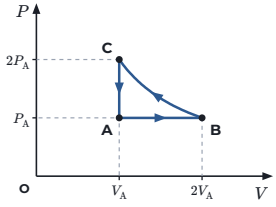
29. For a body cooling according to Newton's law, which of these graphs is a straight line?
(A) $\ln(T - T_s)$ vs t (B) T vs t (C) $(T - T_s)$ vs t (D) $\ln T$ vs t

30. Two solid spheres of the same material, of radii r and $2r$, are heated to the same temperature and kept in the same surroundings. The ratio of their initial rates of fall of temperature (small : large) is:
(A) 2 : 1 (B) 1 : 2 (C) 1 : 4 (D) 4 : 1

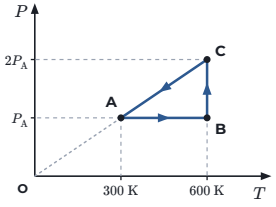
ANSWER KEY Q1–30

1 $\sqrt{2}$ times; 2 times
2 2.5 km/s; 2.74 km/s; v_{rms}
3 500 m/s
4 (D)
5 450 J; 3735 J
6 ≈ 422, 476, 517 m/s; zero
7 450 K (177 °C)
8 (A)
9 (C)
10 Curve (ii): peak at a higher speed and lower (same area)
11 A–r, B–q, C–s, D–p
12 (D)
13 (C)
14 1 : 3
15 (B)
16 Isobaric, isothermal, isochoric; $T_B = 600$ K, $P_C = 2P_A$
17 Isochoric, isobaric, isochoric, isobaric; $2T_0$, $4T_0$, $2T_0$; B and D
18 Isochoric, isothermal, isochoric, isothermal; $2P_A$, P_A, $P_A/2$
19 $T_A = T_B$; no (straight line, not a hyperbola); $\frac{9}{8}T_A$
20 (B)
21 (D)
22 $21N_A$; 4.2
23 T and P; 4 L and 6 L; only T
24 (B)
25 (C)
26 8 min
27 10 °C
28 45 °C; 30 min
29 (A)
30 (A)

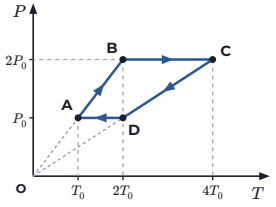
Graph answers · Q16–18



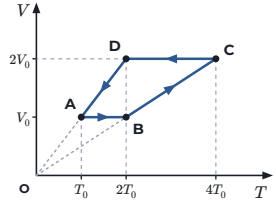
Q16 · P–V



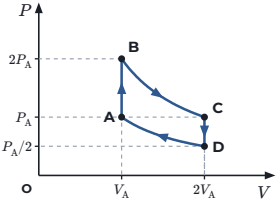
Q16 · P–T



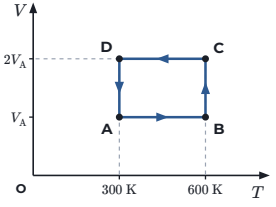
Q17 · P–T



Q17 · V–T



Q18 · P–V



Q18 · V–T